ASSIGNMENT -3-CO3

NAME:	**M.MANASA**
REG NO:	**192425189**
COURSE CODE:	**ITA0329**
COURSE:	**MOBILE COMPUTING**

Q1 - Wireless Network Architecture and Mobility Analysis

Part A - Introduction to Wireless Networks

A wireless network connects devices without using physical cables. It uses radio waves to transfer data between mobile devices and network infrastructure.

1. Characteristics of Wireless Networks

- **Mobility:** Users can move while remaining connected to the network.
- **Limited bandwidth:** Wireless channels usually have less capacity than wired links.
- **Variable signal strength:** Distance, obstacles and interference can affect signal quality.
- **Shared medium:** Multiple devices use the same wireless channel.
- **Handoff:** A mobile device may change from one access point or base station to another.
- **Security:** Wireless communication needs authentication and encryption.

2. Main Components and Their Roles

Component	Role
Mobile node	A moving device such as a smartphone, laptop or ambulance terminal.

Wireless access point / Base station	Provides wireless access and connects mobile nodes to the network.
Communication link	Radio connection used to exchange packets.
Router / Gateway	Forwards packets between networks.
Network server	Provides services, data or applications to users.

3. Wireless Network Architecture

```
Mobile Node 1)))
                      Wireless link
Mobile Node 2))) ---> Access Point / Base Station --->
Router/Gateway ---> Internet/Server
Mobile Node 3)))
```

In this architecture, mobile nodes communicate with an access point or base station. The access point forwards their packets to the router and then to the required server or destination.

4. Communication Constraints

Constraint	**Effect on communication**
Limited bandwidth	Reduces data rate when many devices share the channel.
Weak connectivity	May cause disconnection or retransmission.
Mobility	Can change the point of attachment and interrupt a session.
Interference	Can increase errors and packet loss.

Part B - Wireless Mobility Scenario

Scenario: A user is moving from the coverage area of Access Point A to the coverage area of Access Point B while making a video call.

1. The mobile device is initially connected to Access Point A.

2. As the user moves away, the signal from A becomes weak.

3. The device detects Access Point B and checks its signal quality.

4. If B provides a suitable signal, the device performs handoff and connects to B.

5. Communication continues through B. During the handoff, a short delay or packet loss may occur.

Successful and Interrupted Communication

Case	**Behaviour**
Successful handoff	The new access point is detected and connection is transferred quickly. Most packets reach the destination.
Interrupted handoff	The new network is not found or registration takes too long. Packets may be delayed or lost.
High mobility	The device changes coverage areas frequently, causing repeated handoffs and more interruptions.

Performance measures: packet delivery ratio = (packets received / packets sent) × 100. End-to-end delay is the time taken by a packet to reach its destination. Packet loss is the number of packets that do not reach the destination.

5. Radio Wave Propagation and Signal Degradation Mechanisms

In wireless mobile computing, the physical transmission medium is unguided electromagnetic spectrum. Unlike wired twisted-pair or fiber-optic cables where signals are shielded from external noise, wireless signals experience complex physical phenomena as they propagate from the transmitter to the receiver:

- **Free-Space and Log-Distance Path Loss:** Even in an ideal line-of-sight environment with no obstacles, signal power attenuates proportionally to the square of the distance (d^2) and the square of the carrier frequency (f^2). According to the Friis free-space transmission equation, $Pr = Pt \times Gt \times Gr \times (\lambda / 4\pi d)^2$, doubling the separation distance between a mobile terminal and an Access Point reduces the received signal power by 6 dB (a factor of four). In indoor or urban environments, the path-loss exponent typically increases from 2 to between 3.5 and 5 due to wall penetration and ground reflections.
- **Multipath Propagation and Small-Scale Fading:** Radio waves reflect off buildings, metallic surfaces, vehicles, and terrain, diffract around sharp architectural edges, and scatter off irregular objects. Consequently, multiple copies of the same transmitted signal arrive at the mobile receiver via distinct spatial paths at slightly different times. When waves arrive in phase, constructive interference boosts signal strength; when waves arrive 180 degrees out of phase, destructive interference causes deep fading, producing sudden drops in Signal-to-Noise Ratio (SNR) and bursts of bit errors.
- **Delay Spread and Inter-Symbol Interference (ISI):** Because reflected indirect paths are longer than the direct Line-of-Sight (LoS) path, delayed replicas of a previous symbol can overlap with the arrival of a subsequent symbol at high data rates. This time dispersion corrupts symbol demodulation unless adaptive equalizers or Orthogonal Frequency-Division Multiplexing (OFDM) guard intervals are employed.

- **Shadowing (Slow Fading):** Large physical obstructions such as reinforced concrete structures, hills, or dense foliage absorb and attenuate radio energy, creating coverage shadows or dead zones where the received signal drops below the receiver sensitivity threshold.
- **Doppler Shift and Channel Coherence Time:** When a mobile node moves at high velocity (such as inside a commuter train or emergency ambulance), the relative motion between transmitter and receiver induces a carrier frequency shift $\Delta f = (v \times fc / c) \times \cos(\theta)$. High Doppler spread reduces channel coherence time, requiring rapid channel estimation updates.

6. Medium Access Control (MAC) Challenges in Wireless Networks

Because wireless nodes share a single broadcast radio frequency channel, simultaneous transmissions by neighboring nodes result in frame collisions. Standard wired collision detection (CSMA/CD) cannot operate in wireless networks because a radio transceiver cannot simultaneously transmit and listen on the same frequency band (its own strong transmission saturates its receiver front-end), and spatial attenuation prevents nodes from hearing all other active transmitters.

MAC Challenge	Technical Root Cause	Impact on Mobile Communication	Mitigation Mechanism
Hidden Terminal Problem	Node A and Node C are outside each other's carrier-sensing range, but both lie within range of Access Point B. Both sense the channel as idle and transmit to B simultaneously.	Repeated frame collisions occur at Access Point B, causing throughput collapse under moderate-to-high traffic loads.	RTS/CTS (Request to Send / Clear to Send) virtual carrier sensing using the Network Allocation Vector (NAV).

Exposed Terminal Problem	Node B is transmitting to Node A. Node C (within range of B, but outside range of A) wants to transmit to Node D. C senses B's carrier and defers.	Spatial frequency reuse is wasted, reducing aggregate network capacity even though C's transmission would not interfere at A.	Directional antennas, transmit power control, and dual-channel busy-tone MAC protocols.
Near-Far Effect	A mobile node close to the Access Point transmits at maximum power, drowning out the weak signal arriving from a cell-edge mobile node.	Distant mobile nodes suffer severe packet loss and channel starvation.	Closed-loop Transmit Power Control (TPC) dynamically scaling node output power.
Capture Effect	When two frames overlap at a receiver, the stronger signal can still be decoded if its power exceeds the weaker signal by the capture ratio.	Introduces throughput unfairness favoring nodes geographically close to the Access Point.	Adaptive contention window scaling and fair queuing at the MAC scheduler.

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance) Operational Sequence:

1. Physical and Virtual Carrier Sensing: Before initiating a transmission, the mobile wireless interface inspects physical energy on the radio channel and checks its local Network Allocation Vector (NAV), which records remaining transmission durations announced in headers of overheard frames.

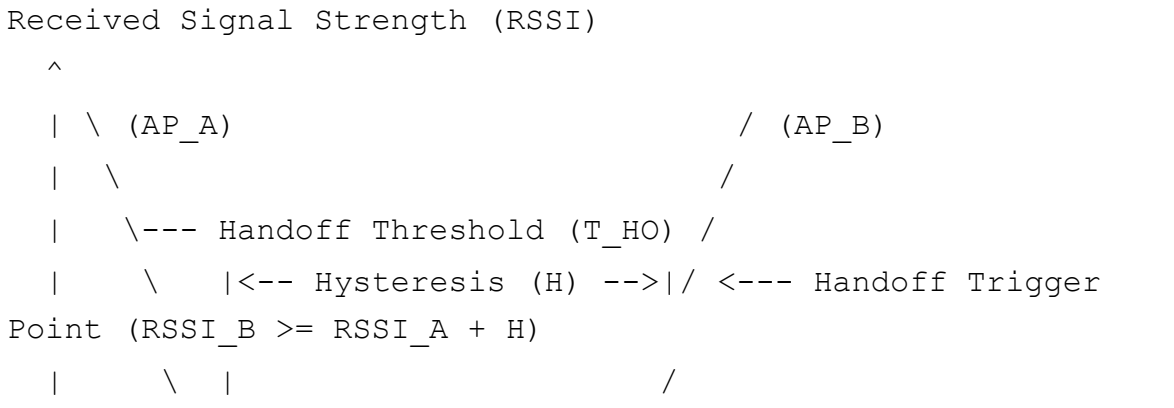
2. DIFS Inter-Frame Waiting Period: If both physical and virtual carrier sense indicators confirm the channel is idle, the node waits for a Distributed Inter-Frame Space (DIFS) duration before transmitting.

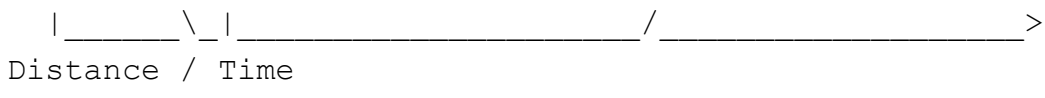
3. Slotted Random Backoff: If the channel is sensed busy, the node defers until the ongoing transmission ends, waits for DIFS, and initializes a random backoff counter uniformly selected from [0, CW], where CW is the Contention Window. The counter decrements by one for each idle slot time and freezes whenever the medium becomes busy.

4. Positive Acknowledgment and Binary Exponential Backoff: Upon error-free reception of a data frame, the destination waits a Short Inter-Frame Space (SIFS) and transmits an ACK frame. If the sender does not receive an ACK within the timeout interval, it doubles its Contention Window ($CW = 2 \times CW + 1$) up to CWmax and schedules a retransmission.

7. Handoff Execution Phases, Hysteresis Margins, and Ping-Pong Prevention

In Part B, as the mobile user moves between Access Point A (AP_A) and Access Point B (AP_B) during an active video call, signal fluctuations near the cell boundary can trigger rapid, repeated switching between the two Access Points—a phenomenon known as the Ping-Pong Effect. To ensure stable handoffs, wireless stations enforce strict signal thresholds and hysteresis margins:





- **Phase 1 - Handoff Initiation and Scanning:** When the average RSSI from AP_A falls below a predefined scanning threshold, the mobile device searches neighboring channels for candidate Access Points. In Passive Scanning, the node listens for periodic Beacon frames (typically broadcast every 100 ms), conserving battery at the cost of higher discovery latency. In Active Scanning, the node actively transmits Probe Request frames across channels and collects Probe Responses, completing discovery within 20 to 40 ms.
- **Phase 2 - Hysteresis-Based Handoff Decision:** Rather than triggering handoff the instant RSSI_B exceeds RSSI_A, the station requires RSSI_B ≥ RSSI_A + H, where H is the Hysteresis Margin (typically 3 dB to 6 dB), sustained over a Time-to-Trigger (TTT) timer. This filters out transient multipath fades and prevents ping-pong oscillations.
- **Phase 3 - Handoff Execution (Hard vs. Soft Handoff):** In IEEE 802.11 Wi-Fi networks, stations typically execute a Hard Handoff (Break-Before-Make), disassociating from AP_A before completing re-association and 802.11i/WPA2/WPA3 key negotiation with AP_B. In contrast, cellular architectures often support Soft Handoff (Make-Before-Break), maintaining simultaneous radio links with both base stations until the target link stabilizes, eliminating real-time video frame drops.

8. IEEE 802.11 Physical Layer Modulation, MIMO, and Fast BSS Transition (802.11r)

To sustain high-definition video calls and real-time mobile applications over fluctuating wireless links, modern WLAN architectures combine adaptive modulation schemes with fast handoff amendments:

- **Adaptive Modulation and Coding (AMC):** Wireless transceivers continuously monitor Signal-to-Noise Ratio (SNR) and dynamically select an appropriate Modulation and Coding Scheme (MCS). When the mobile device is close to an Access Point with high SNR, it uses dense constellations such as 64-QAM or 256-QAM with a 5/6 coding rate to achieve maximum throughput. As the user walks toward the cell edge and SNR drops, the radio steps down to 16-QAM, QPSK, or robust BPSK with 1/2 rate convolutional/LDPC coding to preserve link reliability.
- **MIMO Spatial Diversity and Multiplexing:** Multiple-Input Multiple-Output (MIMO) technology utilizes multiple antennas at both the Access Point and mobile terminal to exploit spatial multipath propagation. Spatial Multiplexing transmits independent data streams simultaneously across parallel spatial channels to multiply throughput, whereas Maximal Ratio Combining (MRC) diversity combines signals from multiple antennas to combat deep fading.
- **IEEE 802.11r Fast BSS Transition:** Standard WPA2/WPA3 enterprise authentication requires a full 802.1X/EAP exchange with a backend RADIUS server upon every Access Point transition, introducing 200 to 500 ms of audio/video interruption. IEEE 802.11r (Fast BSS Transition) allows the mobile station to pre-derive cryptographic keys with the target Access Point while still connected to the current Access Point, reducing Layer 2 handoff interruption to under 50 ms.

9. Transport Layer Impact: Why Standard TCP Mistreats Wireless Packet Loss

In traditional wired networks, packet loss almost exclusively results from router buffer overflow caused by network congestion. Consequently, standard Transmission Control Protocol (TCP) congestion control algorithms (such as TCP Reno or CUBIC) respond to every duplicate acknowledgment or retransmission timeout by halving the congestion window (cwnd) and

entering slow start. In a wireless mobile environment, however, packet loss frequently stems from transient radio interference, multipath fading, or brief handoff disconnections rather than congestion. When TCP halves its transmission rate during a wireless fade, link utilization collapses even after the radio signal recovers. To mitigate this, mobile systems employ Split-TCP proxies (Indirect-TCP) at the base station, Snoop TCP agents that cache and locally retransmit lost wireless segments, and Explicit Congestion Notification (ECN) to distinguish true router congestion from wireless bit errors.

Conclusion: Wireless networks support communication while users move, but bandwidth limits, signal changes and handoff delays can affect performance. Good coverage and fast handoff improve communication.

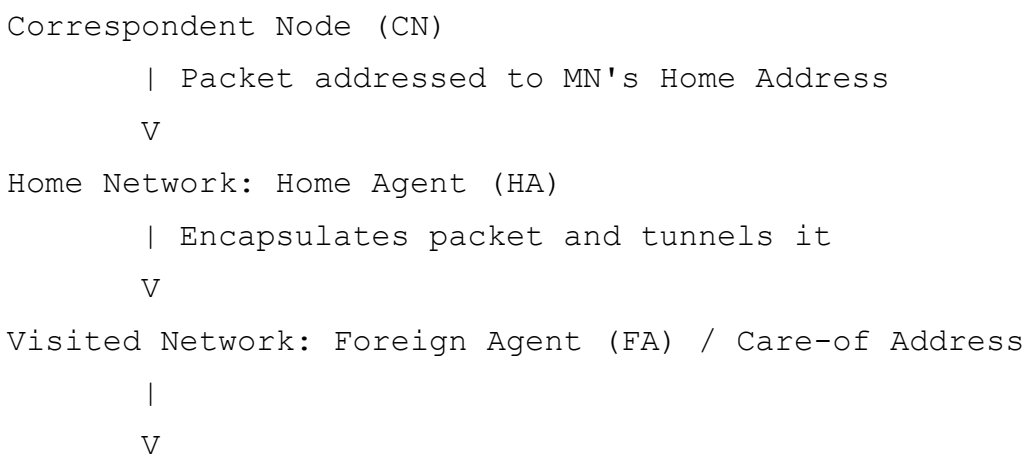
Q2 - Mobile IP and Mobility Management

Mobile IP allows a mobile node to move from one network to another while keeping its permanent home IP address. It helps maintain ongoing communication even when the node changes its network attachment.

1. Mobile IP Components

Component	Meaning/Function
Mobile Node (MN)	The device that moves between networks.
Home Agent (HA)	A router on the home network that stores the mobile node's current location and forwards packets.
Foreign Agent (FA)	A router in a visited network that assists the mobile node in traditional Mobile IPv4.
Correspondent Node (CN)	The device or server communicating with the mobile node.
Care-of Address (CoA)	The temporary address representing the mobile node's current location.

2. Mobile IP Architecture



```
Mobile Node (MN)
```

The mobile node keeps its home address. When it enters a visited network, it obtains a care-of address and registers it with the Home Agent.

3. Registration and Location Updating

1. The mobile node detects that it is in a visited network.

2. It discovers a Foreign Agent or obtains a care-of address.

3. The mobile node sends a registration request containing its home address and care-of address to the Home Agent.

4. The Home Agent authenticates and checks the request.

5. If accepted, the Home Agent stores the binding between the home address and care-of address.

6. The Home Agent sends a registration reply.

4. Packet Delivery and Tunnelling

1. The Correspondent Node sends a packet to the mobile node's permanent home address.

2. The packet reaches the home network and is intercepted by the Home Agent.

3. The Home Agent encapsulates the original packet inside another packet and sends it through a tunnel to the care-of address.

4. The tunnel endpoint decapsulates the packet and delivers the original packet to the mobile node.

5. Important Mobile IP Challenges

Challenge	Explanation / Possible improvement
Registration delay	Registration takes time after movement. Reduce unnecessary registrations and use efficient agent discovery.
Tunnelling overhead	Extra headers increase packet size. Use efficient encapsulation.
Triangular routing	Packets may travel CN -> HA -> MN instead of directly to MN. Route optimisation can reduce the indirect path where supported.
Packet loss	Packets may be lost during handoff. Buffer packets or retransmit when appropriate.
Security	False registration can redirect traffic. Authenticate registration messages and protect mobility information.
Scalability	Many mobile nodes create signalling load. Use efficient registration and forwarding methods.

6. Agent Discovery Protocol Mechanics and Movement Detection

Before a Mobile Node (MN) can register a new Care-of Address, it must determine whether it is attached to its home subnet or a foreign subnet. Mobile IP extends standard ICMP Router Discovery (RFC 1256) with two control messages:

- **Agent Advertisement Messages:** Home Agents and Foreign Agents periodically broadcast ICMP Router Advertisement messages appended with a Mobility Agent Advertisement Extension. This extension carries the agent's Care-of Address(es), registration lifetime limit, and capability flags (such as H-bit for Home Agent, F-bit for Foreign Agent, R-bit indicating

registration with a Foreign Agent is mandatory even when using a co-located CoA, and encapsulation support bits for GRE or Minimal Encapsulation).

- **Agent Solicitation Messages:** If a Mobile Node arrives on a new link and needs immediate connectivity without waiting for the next periodic advertisement, it broadcasts an ICMP Router Solicitation message (TTL = 1) to prompt local mobility agents to transmit an immediate Agent Advertisement.
- **Movement Detection Algorithms:** Mobile Nodes employ two primary movement detection algorithms: (a) Lifetime Expiration, where the MN assumes it has moved if it fails to receive a new advertisement from its current agent before the previous advertisement's lifetime expires; and (b) Network Prefix Comparison, where the MN compares the network prefix of newly received Agent Advertisements against its current CoA prefix, immediately triggering registration upon detecting a prefix mismatch.

7. IP Tunnelling and Encapsulation Formats

When the Home Agent intercepts a datagram destined for the MN's Home Address (using Gratuitous ARP and Proxy ARP on the home link so local routers forward the MN's traffic to the HA), it encapsulates the datagram using one of three standardized encapsulation schemes:

Encapsulation Scheme	**Header Overhead**	**Header Structure & Fields**	**Operational Trade-offs**
IP-in-IP Encapsulation (RFC 2003)	20 bytes (Full Outer IPv4 Header)	Outer IP Header (Src = HA, Dst = CoA, Protocol = 4) + Inner Original IP Header (Src = CN, Dst = MN Home Addr) + Payload.	Mandatory default in Mobile IP; simple to process but duplicates redundant fields (TTL, ToS, Identification) increasing bandwidth

			overhead.
Minimal Encapsulation (RFC 2004)	8 to 12 bytes	Outer IP Header (Protocol = 55) + Minimal Forwarding Header (Original Dst, optional Original Src) + Payload.	Reduces byte overhead by eliminating duplicated IP header fields, but cannot be used if the original datagram is already fragmented.
Generic Routing Encapsulation (GRE - RFC 2784)	24 bytes (Outer IP + 4-byte GRE Header)	Outer IP Header (Protocol = 47) + GRE Header (Flags, Protocol Type) + Inner IP Packet.	Supports multi-protocol payloads and optional sequence numbering/keying, at the cost of higher header overhead.

8. Route Optimization and Mobile IPv6 (MIPv6) Architectural Improvements

In standard Mobile IPv4, packets from a Correspondent Node (CN) to the MN follow a suboptimal triangular path (CN -> HA -> MN), whereas return packets from the MN to the CN travel directly (MN -> CN) unless reverse tunnelling is enforced to satisfy ingress filtering firewalls. Triangular routing increases end-to-end latency and creates a traffic bottleneck at the Home Agent.

- **Route Optimization Extensions:** Route Optimization allows the Home Agent to transmit a Binding Update message directly to the Correspondent Node, informing the CN of the MN's current Care-of Address. The CN caches this binding in its Binding Cache and tunnels subsequent packets directly to the CoA, eliminating the Home Agent from the forwarding

path. When the MN moves again, the previous Foreign Agent can forward in-flight packets to the new CoA using Smooth Handoff binding notifications.

- **Mobile IPv6 (MIPv6) Enhancements:** Mobile IPv6 integrates mobility natively into the IPv6 protocol stack without requiring Foreign Agents. Using IPv6 Stateless Address Autoconfiguration (SLAAC) or DHCPv6, the MN automatically configures a globally routable Co-located Care-of Address on every visited network. Furthermore, MIPv6 replaces IP-in-IP encapsulation for Route Optimization with an IPv6 Type 2 Routing Header (for HA-to-MN/CN-to-MN packets) and a Home Address Destination Option (for MN-to-CN packets), reducing header overhead while verifying binding authenticity via the Return Routability Procedure.

9. Reverse Tunnelling, Proxy ARP, and Ingress Filtering Traversal

Two critical low-level networking mechanisms ensure seamless packet interception and firewall traversal in Mobile IP:

- **Proxy ARP and Gratuitous ARP on the Home Link:** When the Mobile Node leaves its home subnet, other hosts and routers on the home LAN still attempt to resolve the MN's permanent Home Address into a MAC address using Address Resolution Protocol (ARP). The Home Agent answers these ARP requests on behalf of the absent MN using Proxy ARP, and broadcasts Gratuitous ARP frames upon registration so all local nodes immediately update their ARP caches to point to the Home Agent's MAC address. When the MN eventually returns home, it broadcasts its own Gratuitous ARP to reclaim its traffic.
- **Reverse Tunnelling for Ingress Filtering Compliance:** Modern enterprise and ISP edge routers implement Ingress Filtering (RFC 2827) to block IP spoofing attacks by dropping outgoing packets whose source IP address does not match the topological subnet prefix of the local access network. When a Mobile Node in a foreign network sends packets directly to a

Correspondent Node using its permanent Home Address as the packet source address, the foreign router drops those packets. To overcome ingress filtering, Mobile IP supports Reverse Tunnelling (RFC 3024), wherein the Mobile Node encapsulates outgoing packets and tunnels them back to its Home Agent (MN -> FA -> HA -> CN), ensuring topologically valid source addresses on the visited network.

10. Joint Answer: Frequent Network Changes During an Active Session

Frequent movement may cause repeated agent discovery, registration delay, changing care-of addresses, packet loss and additional tunnelling overhead. The session may experience delay or temporary interruption. These problems can be reduced by fast agent discovery, timely registration, buffering packets during handoff, selecting a strong target network, authenticating registration messages and reducing unnecessary signalling.

Conclusion: Mobile IP supports location-independent communication by registering a care-of address and tunnelling packets through the Home Agent. Its main concerns are delay, overhead, packet loss, security and scalability.

Q3 - Cellular Networks versus Ad Hoc Wireless Networks

1. Cellular Network

A cellular network is an infrastructure-based wireless network. A geographical area is divided into cells. Each cell is served by a base station. Mobile devices communicate through the base station, which connects them to the wider network.

```
Mobile Device))) Base Station / Cell Tower ---> Core Network
---> Internet
```

- Base stations manage wireless access and support handoff between cells.
- Communication is mainly infrastructure-assisted and centrally managed.
- It provides wide-area coverage when the required infrastructure is available.

2. Ad Hoc Wireless Network

An ad hoc wireless network is an infrastructure-less network formed by mobile nodes. It does not require a fixed base station or access point. Nodes communicate directly when they are in range, or through other nodes using multi-hop communication.

```
Node A <----> Node B <----> Node C
                |
                V
              Node D
```

- Each node can work as both a host and a router.
- The topology changes when nodes move or links fail.
- It can be deployed quickly in places where fixed infrastructure is unavailable.

3. Comparison

Parameter	Cellular Network	Ad Hoc Network
Infrastructure	Uses base stations and fixed infrastructure.	Does not require fixed infrastructure.
Communication	Usually through a base station.	Direct or multi-hop between mobile nodes.
Routing	Infrastructure-assisted routing.	Distributed routing between nodes.
Mobility	Uses cellular mobility and handoff management.	Routes must adapt to moving nodes.
Topology	Planned cell structure.	Dynamic topology.
Scalability	Can support many users with planned capacity.	Depends on node density, wireless links and routing overhead.
Handoff	Handoff between cells/base stations.	Routes or neighbor links are updated as nodes move.
Failure handling	Base-station or core-network failure can affect service.	Alternative peer paths may be used if available.
Applications	Mobile voice, broadband and public cellular service.	Disaster recovery, military, temporary and field networks.

4. Cellular Architecture Internals: Frequency Reuse and Capacity Scaling

Because electromagnetic radio spectrum is strictly regulated and scarce, cellular networks maximize spectral efficiency and user capacity across wide geographic regions through structured spatial engineering:

- **Frequency Reuse and Cluster Geometry:** A large service region is tessellated into hexagonal cells, and the total available pool of radio channels (S) is partitioned into N disjoint channel groups (cluster size $N = i^2 + ij + j^2$, such as N = 3, 4, or 7). Cells separated by a minimum reuse distance $D = R \times \sqrt{(3N)}$ (where R is the cell radius) simultaneously reuse the exact same carrier frequencies without mutual destruction. Repeating a cluster M times multiplies total system capacity to $C = M \times S$.
- **Co-Channel and Adjacent-Channel Interference Management:** Two cells using identical frequency channels are termed co-channel cells. The Signal-to-Interference Ratio (SIR) at a mobile receiver is approximated by SIR = (D/R)^n / i_0, where n is the path-loss exponent and i_0 is the number of first-tier co-channel interferers (typically 6). Increasing the cluster size N reduces co-channel interference but lowers channel density per cell.
- **Cell Splitting and Antenna Sectoring:** To accommodate dense urban user populations without acquiring additional spectrum, operators employ two capacity-expansion techniques: (a) Cell Splitting, which subdivides congested macro-cells into smaller micro-cells or pico-cells with reduced base-station transmit power, increasing spatial reuse; and (b) Cell Sectoring, which replaces an omnidirectional base-station antenna with directional antennas (typically three 120° sectors or six 60° sectors), reducing the number of co-channel interferers and boosting SIR.
- **Core Network Mobility Databases:** Cellular mobility is centrally coordinated by core network databases and controllers—such as the Home Location Register (HLR) / Home Subscriber Server (HSS) storing permanent subscriber profiles, the Visitor Location Register (VLR) / Mobility Management Entity (MME) tracking current cell/tracking-area locations, and Base Station Controllers (BSC) / eNodeBs executing measurement-driven intra-cell and inter-cell handovers.

5. Mobile Ad Hoc Network (MANET) Operational Challenges

Unlike cellular networks where single-hop radio links connect terminals to high-power, mains-fed base stations, Mobile Ad Hoc Networks (MANETs) operate over peer-to-peer multi-hop wireless paths that introduce distinct architectural complexities:

- **Multi-Hop Spatial Contention and Throughput Degradation:** In a multi-hop string topology (Node A -> Node B -> Node C -> Node D), adjacent links share the same broadcast spectrum. When Node B transmits to Node C, Node A cannot simultaneously transmit to Node B without causing a collision at B, and Node D cannot transmit due to interference. Consequently, end-to-end throughput degrades sharply as hop count increases (proportional to 1/k for k hops).
- **Asymmetric and Unidirectional Wireless Links:** Differences in battery levels, antenna gains, or local noise floors among heterogeneous mobile devices can produce unidirectional (asymmetric) wireless links where Node A can hear Node B, but Node B cannot hear Node A. Routing protocols that assume bidirectional links fail unless explicit link-symmetry checks are enforced.
- **Energy Constraints and Network Partitioning:** Because portable mobile nodes rely on finite battery capacity, acting as an intermediate relay router to forward third-party traffic drains energy. Without energy-aware routing metrics, centrally located nodes rapidly exhaust their batteries, causing network partitioning.

6. Hybrid Cellular-Ad Hoc Relaying Architectures (iCAR and Multi-Hop Cellular)

Modern wireless systems increasingly bridge the divide between purely infrastructure-based cellular networks and purely infrastructure-less ad hoc networks by deploying hybrid architectures:

- **Load Balancing via Ad Hoc Relaying Stations:** Integrated Cellular and Ad Hoc Relaying Systems (iCAR) place portable Ad Hoc Relaying Stations (ARS) near cell boundaries to divert traffic from congested hot-spot cells into neighboring lightly loaded cells via multi-hop ad hoc links, dynamically balancing cellular load without adding new base-station towers.
- **5G/LTE Device-to-Device (D2D) Proximity Services:** Device-to-Device (D2D) communication in LTE-Advanced and 5G NR allows proximate mobile devices to establish direct peer-to-peer radio links under the control and synchronization of the cellular base station (gNodeB), extending cell-edge coverage during partial infrastructure outages.

7. Evolution Across Cellular Generations (2G to 5G) vs. Specialized MANET Variants

To contextualize infrastructure dependency, cellular networks have evolved from circuit-switched 2G (GSM/TDMA) and packet-switched 3G (UMTS/HSPA) to all-IP flat architectures in 4G LTE (EPC) and cloud-native service-based 5G NR cores supporting ultra-reliable low-latency communication (URLLC). Meanwhile, ad hoc wireless networks have specialized into domain-specific paradigms: Vehicular Ad Hoc Networks (VANETs) utilizing IEEE 802.11p/C-V2X for high-speed roadside safety telemetry, Wireless Sensor Networks (WSNs) optimized for ultra-low-power environmental monitoring, and Flying Ad Hoc Networks (FANETs) deploying autonomous UAV drones as aerial relays over disaster zones.

8. Joint Engineering Decision - Disaster Environment

For an emergency or disaster area where fixed infrastructure may be unavailable, an ad hoc wireless network is a suitable choice. Rescue teams, police and ambulances can form a temporary network and forward messages through nearby nodes. Multi-hop communication can extend coverage beyond the range of one device. If some nodes or links fail, the routing protocol can search for another available path.

Limitation: An ad hoc network may have limited bandwidth, changing routes and battery constraints. If cellular infrastructure is available and reliable, it can provide wider coverage and managed service.

Conclusion: Cellular networks are suitable when infrastructure and broad coverage are available. Ad hoc networks are useful for rapid communication without fixed infrastructure.

Q4 - Applications of Mobile IP and Routing Requirements

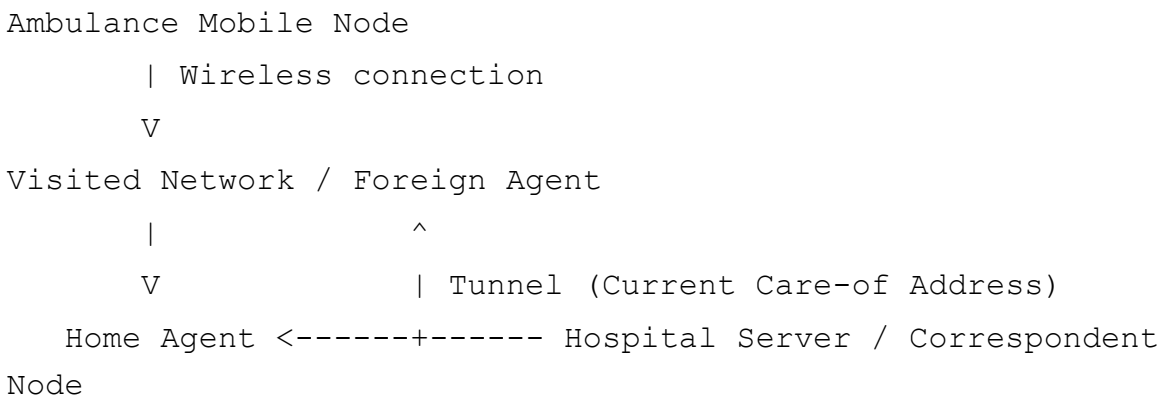
Selected Application: Emergency Ambulance Communication

Mobile IP can be used in an emergency healthcare system where ambulances move between different network areas but must remain connected to hospitals.

1. Network Entities

- **Mobile Node:** Ambulance communication device or onboard computer.
- **Home Agent:** Router that maintains the ambulance's home-address-to-care-of-address binding.
- **Foreign Agent / visited network:** Network currently serving the ambulance.
- **Correspondent Node:** Hospital server or emergency control centre.

2. Application Architecture



The ambulance sends patient details, location and emergency updates to the hospital. When it enters another network, it obtains a new care-of address and registers it with the Home Agent. Packets can then be tunnelled to its current location.

3. Communication Flow

1. The ambulance device connects to the available wireless network.

2. It discovers the visited network and obtains a care-of address.

3. It registers the care-of address with the Home Agent.

4. The hospital server sends information to the ambulance's home address.

5. The Home Agent tunnels packets to the current care-of address.

6. The ambulance receives the information and continues sending updates.

4. Routing and Performance Requirements

Requirement	**Why it is important**
Low delay	Emergency instructions and patient information must arrive quickly.
High packet delivery	Important medical data should reach the hospital correctly.
Continuous connectivity	Communication should continue while the ambulance moves.
Fast handoff	Network changes should cause minimum interruption.
Security	Patient and emergency information must be protected.
Reliability	The system should detect failures and use an available connection or route.

5. Telemedicine Data Traffic Profiles and Quality of Service (QoS) Budgets

An emergency ambulance operating as a mobile telemedicine unit simultaneously transmits heterogeneous medical data streams to the hospital trauma center. Each stream imposes strict Quality of Service (QoS) constraints on the underlying Mobile IP transport layer:

Telemedicine Traffic Stream	**Data Rate / Bandwidth**	**Max End-to-End Latency**	**Max Packet Loss Tolerance**	**Clinical Consequence of Degradation**
12-Lead Real-Time ECG & Vital Signs (SpO2, NIBP)	12 – 64 kbps (continuous)	< 150 ms	< 0.1% (near zero)	Waveform distortion or missed arrhythmia alerts during critical cardiac transport.
Paramedic-Physician Full-Duplex VoIP Audio	32 – 64 kbps (G.711 / AMR-WB)	< 150 ms (Jitter < 30 ms)	< 1.0%	Unintelligible voice instructions during emergency airway or trauma procedures.
Live Ambulance Cabin Video Consultation	512 kbps – 2.5 Mbps (H.264/H.265)	< 250 ms	< 1.0%	Frame freezing or macro-blocking obscuring visual patient triage assessment.
High-Resolution Ultrasound / Imaging Transfer	2 – 10 Mbps (burst)	< 2 – 5 seconds	0% (TCP reliable)	Delayed diagnostic preparation at the hospital

				emergency department.

6. Minimizing Handoff Latency: Hierarchical and Fast Mobile IP Extensions

When an ambulance travels at 80 to 110 km/h across municipal Wi-Fi hotspots, roadside infrastructure, and cellular gateways, standard Mobile IP registration round-trip time to a distant Home Agent can exceed 300–500 ms, causing unacceptable interruptions to live ECG and VoIP streams. Two architectural extensions address this limitation:

- **Hierarchical Mobile IPv6 (HMIPv6):** HMIPv6 introduces a local Mobility Anchor Point (MAP) within the visited regional network that acts as a local Home Agent. The ambulance configures two Care-of Addresses: a Regional Care-of Address (RCoA) on the MAP's subnet and an On-Link Care-of Address (LCoA) on its current access router. When the ambulance moves between access routers within the same metropolitan region, it only updates its LCoA with the nearby MAP without signalling the distant Home Agent, reducing handover latency and global signalling load.
- **Fast Handovers for Mobile IPv6 (FMIPv6):** FMIPv6 uses Link-Layer (Layer 2) triggers—such as rapid SNR decay—to anticipate an imminent handover before Layer 3 connectivity breaks. In Predictive Mode, the ambulance pre-configures its New Care-of Address (NCoA) on the target access router via the current access router and establishes a temporary bidirectional tunnel between the Previous Access Router (PAR) and New Access Router (NAR). Incoming packets are buffered at the NAR and delivered immediately upon radio attachment, reducing packet loss to near zero.

7. Security Architecture for Mobile Healthcare Tunnels

Transmitting sensitive patient medical records (protected under healthcare privacy standards) over public wireless links exposes the system to eavesdropping, spoofing, and registration hijacking:

- **Registration Message Authentication:** Every Mobile IP Registration Request includes a Mobile-Home Authentication Extension computed using HMAC-SHA256 with a shared secret key and a replay-protection timestamp or nonce. This prevents attackers from forging Care-of Address updates to hijack hospital traffic.
- **IPsec ESP Tunnel Encryption and AAA Integration:** To protect patient confidentiality and data integrity across untrusted foreign networks, the Mobile IP tunnel is encapsulated inside an IPsec Encapsulating Security Payload (ESP) tunnel in transport or tunnel mode, paired with AAA (Authentication, Authorization, and Accounting) servers (RADIUS/Diameter) for dynamic session key distribution.

8. Multi-Homed Ambulance Interfaces and Vertical Handover Management

To achieve ultra-high reliability during critical patient transport, modern emergency ambulances carry multi-homed mobile routers equipped with simultaneous 5G/LTE cellular modems, municipal IEEE 802.11p/ax roadside transceivers, and Low-Earth-Orbit (LEO) satellite terminals. Using Network Mobility (NEMO Basic Support - RFC 3963) and Multiple Care-of Addresses Registration (RFC 5648), the ambulance's mobile router registers multiple concurrent Care-of Addresses with its Home Agent. This enables seamless Vertical Handover across heterogeneous access technologies (e.g., switching from municipal Wi-Fi to 5G cellular) and bicasting of critical ECG telemetry over two independent links simultaneously.

9. Suitability of Mobile IP

Mobile IP is suitable because it allows the ambulance to keep its permanent home address while moving between networks. It supports communication with the hospital without requiring the correspondent to know the ambulance's changing location. However, registration delay, packet loss and tunnelling overhead must be managed.

Conclusion: Mobile IP can support mobile healthcare and ambulance communication by maintaining a binding to the current location and forwarding packets through a tunnel.

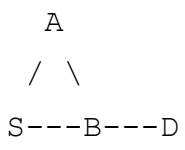
Q5 - Proactive Routing Protocols in Mobile Networks

A proactive routing protocol maintains routes to destinations before data is sent. Nodes periodically exchange routing information and update their routing tables. Therefore, a route may already be available when an application needs to send a packet.

1. DSDV Protocol

Destination-Sequenced Distance Vector (DSDV) is a proactive routing protocol. It uses destination sequence numbers to identify fresh routes and reduce routing loops. Each routing-table entry contains destination, next hop, hop count and sequence number.

2. Sample Network Topology



Example: Node S can reach destination D through B. If the S-B link fails, the network exchanges updated routing information and searches for a valid route through another neighbor if one exists.

3. DSDV Route Table Example

Node	Destination	Next Hop	Hop Count	Sequence No.	Status
S	D	B	2	20	VALID
B	D	D	1	20	VALID
A	D	B	2	20	VALID

After a topology change, a node receives a newer sequence number or a route with a better metric. It updates the routing table with the fresher or better route. Invalid or expired entries are removed.

4. Simplified DSDV Route Update Pseudocode

```
ALGORITHM UpdateDSDVRoute(Node, ReceivedRoute)
INPUT: Destination, NextHop, HopCount, SequenceNumber
OUTPUT: Updated routing table
BEGIN
    Destination <- ReceivedRoute.Destination
    NewHopCount <- ReceivedRoute.HopCount + 1
    IF Destination is not in RouteTable THEN
        ADD the received route
    ELSE IF ReceivedSequenceNumber >
RouteTable[Destination].SequenceNumber THEN
        REPLACE the old route
    ELSE IF ReceivedSequenceNumber ==
RouteTable[Destination].SequenceNumber AND NewHopCount <
RouteTable[Destination].HopCount THEN
        REPLACE the old route
    END IF
    RETURN RouteTable
END
```

5. Performance Analysis

Parameter	**DSDV behaviour**
Route availability	Routes are maintained before data transmission.
Initial route delay	Usually low because routes are already

	present.
Control overhead	Periodic updates consume bandwidth.
Convergence	After topology changes, nodes need time to receive and process updates.
Mobility	Frequent movement causes more routing-table changes and updates.
Network size	Larger networks can increase the amount of routing information and control traffic.

6. DSDV Update Mechanisms: Full Dumps, Incremental Updates, and Damping Fluctuations

To balance routing table accuracy against limited wireless bandwidth, DSDV regulates how and when routing updates are broadcast across the network:

- **Periodic Full Dumps vs. Incremental Updates:** Every destination node periodically increments its own sequence number by an even integer (e.g., 20 -> 22 -> 24) and broadcasts its routing table. To minimize control traffic, DSDV distinguishes between Full Dumps (which transmit all available routing entries across multiple network protocol data units when major topological shifts occur) and Incremental Updates (which transmit only entries that have changed since the last full dump, fitting inside a single packet).
- **Triggered Updates and Odd Sequence Numbers for Broken Links:** When a mobile node detects that its link to a next-hop neighbor has broken, it immediately sets the hop count for all destinations routed through that neighbor to infinity (∞) and increments the destination's sequence number by 1 to an odd integer (e.g., 20 -> 21). Because odd sequence numbers indicate a broken route, neighboring nodes immediately invalidate stale entries upon

receiving the triggered update, preventing the classic Bellman-Ford 'count-to-infinity' routing loop problem.

- **Damping Route Fluctuations via Settling Time:** In a mobile network, routing updates for the same destination sequence number may arrive via different paths at slightly different times—for example, a longer path with fewer hops might arrive milliseconds before a shorter path that traversed congested intermediate nodes. If a node immediately re-broadcasts every transient metric improvement, control traffic explodes. DSDV solves this by maintaining a Settling Time field in each routing table entry (calculated as a weighted moving average of past convergence times). A node updates its internal forwarding table immediately upon receiving a better metric, but delays broadcasting the triggered update until the settling time expires.

7. Step-by-Step Topology Change Trace in DSDV

Using the sample topology (S connected to A and B; A connected to B; B connected to D), consider the routing table evolution at Node S when link B-D breaks and Node D moves within range of Node A:

Time / Event	**Routing Action**	**S's Entry for D (Next Hop, Metric, Seq#)**	**Network State Explanation**
t = 0 (Initial)	Steady-state operation	(B, 2, 20)	S routes to D via B in 2 hops with even sequence number 20.
t = 1 (Link B-D breaks)	B detects broken link to D	(B, 2, 20) [Pending]	B sets metric to D as ∞ and assigns odd sequence number 21.
t = 2 (Triggered	B broadcasts (D, ∞,	(B, ∞, 21)	S receives odd

Update)	21) to S and A	[INVALID]	sequence number 21 > 20 and marks route to D invalid.
t = 3 (D moves near A)	D establishes link with A and advertises Seq# 22	(B, ∞, 21) [INVALID]	A updates its table for D to (D, 1, 22) and broadcasts update.
t = 4 (Convergence)	S receives A's update (D, 2, 22)	(A, 2, 22) [VALID]	Even sequence number 22 > 21 replaces broken entry; path S->A->D restored.

8. Comparison with Other Proactive Protocols (OLSR and WRP)

While DSDV adapts classical distance-vector routing using sequence numbers, other proactive protocols optimize overhead differently:

- **OLSR (Link-State Proactive):** Optimized Link State Routing (OLSR) reduces flooding overhead by selecting a subset of neighbors as Multipoint Relays (MPRs). Only MPR nodes forward broadcast link-state control messages, significantly improving scalability in dense ad hoc networks compared to DSDV.
- **WRP (Path-Finding Proactive):** Wireless Routing Protocol (WRP) maintains four separate tables (Distance, Routing, Link-Cost, and Message Retransmission List) and requires explicit acknowledgments for update packets to achieve faster loop-free convergence than DSDV, though at the cost of higher memory consumption on mobile devices.

9. Quantitative Overhead Complexity Analysis of Proactive Routing

In a mobile ad hoc network consisting of N mobile nodes with an average node degree d and periodic update interval T_p, every node broadcasts a routing vector containing O(N) destination entries once every T_p seconds. Consequently, the periodic control message complexity scales as O(N) packet transmissions per update cycle, while total control byte overhead across the network scales as $O(N^2)$ per cycle. When node mobility velocity increases and link breakages trigger unscheduled incremental updates at rate λ_{link}, total control overhead becomes $O(N^2 + \lambda_{link} \times N)$, which can saturate low-bandwidth wireless links if N exceeds several hundred mobile terminals.

Conclusion: DSDV is useful when low route-discovery delay is important and the network topology is reasonably stable. Its main disadvantage is periodic control overhead, especially when nodes move frequently.

Q6 - Reactive Routing and Protocol Comparison

A reactive routing protocol discovers a route only when a source node needs to send data and no valid route is available. This is called on-demand routing.

1. AODV Protocol

Ad hoc On-Demand Distance Vector (AODV) is a reactive routing protocol. It discovers routes when required and maintains active routes while they are being used.

2. AODV Control Messages

Message	Purpose
RREQ - Route Request	Broadcast by the source to search for a route to the destination.
RREP - Route Reply	Sent back when the destination or a node with a valid route is found.
RERR - Route Error	Informs affected nodes that a route or link has failed.

3. AODV Route Discovery

1. Source S checks its routing table for a valid route to destination D.

2. If no route exists, S broadcasts a Route Request (RREQ) to its neighbors.

3. Intermediate nodes record a reverse route to S and forward the RREQ if they have not already processed it.

4. When D or an intermediate node with a valid route receives the request, it sends a Route Reply (RREP).

5. The RREP travels back to S and establishes the forward route.

6. S sends data using the discovered route.

4. Route Maintenance

1. Nodes monitor links on an active route.

2. If a link fails, the affected route is marked invalid.

3. If an alternate valid route exists, it is selected.

4. Otherwise, a Route Error (RERR) is sent to the source.

5. If communication is still required, the source starts a new route discovery.

5. AODV Route Discovery Pseudocode

```
ALGORITHM ReactiveRouteDiscovery(Source, Destination)
INPUT: Source and Destination
OUTPUT: Route or ROUTE_NOT_FOUND
BEGIN
    IF a valid route exists THEN RETURN the route
    Create a new Route Request (RREQ)
    Broadcast RREQ to neighbors
    WAIT for RREP or route failure
    IF RREP is received THEN
        Establish reverse and forward paths
        Mark route as VALID
        RETURN the route
    ELSE
        RETURN ROUTE_NOT_FOUND
    END IF
END
```

6. AODV Control Packet Header Fields and Loop Prevention

To prevent duplicate packet storms and guarantee loop-free paths without carrying full source-route lists in every data packet, AODV embeds specific identifiers and sequence counters inside its control messages:

- **RREQ Header Structure & Duplicate Suppression:** Contains Type (1), Flags (J, R, G, D, U), Hop Count, RREQ ID, Destination IP Address, Destination Sequence Number, Originator IP Address, and Originator Sequence Number. The tuple (Originator IP Address, RREQ ID) uniquely identifies each discovery flood; any intermediate node receiving a duplicate (Originator IP, RREQ ID) silently discards it. Furthermore, an intermediate node may only generate a gratuitous RREP if its cached Destination Sequence Number is greater than or equal to the Destination Sequence Number carried in the RREQ, guaranteeing path freshness.
- **RREP Header Structure & Route Lifetime Management:** Contains Type (2), Flags (R, A), Prefix Size, Hop Count, Destination IP Address, Destination Sequence Number, Originator IP Address, and Route Lifetime (in milliseconds). As the unicast RREP traverses the reverse path back to the source, each node creates a forward routing table entry pointing to the neighbor from which it received the RREP and resets the route's Active Route Timeout timer.
- **RERR Header Structure & Precursor Lists:** Contains Type (3), Flag (N - No Delete), DestCount, and a list of (Unreachable Destination IP Address, Unreachable Destination Sequence Number) pairs. Nodes maintain a Precursor List of upstream neighbors that actively forward packets through them; when a downstream link breaks, RERR is propagated specifically to affected precursors.

7. Expanding Ring Search (ERS) and Local Route Repair Optimizations

Network-wide flooding of RREQ packets consumes significant wireless bandwidth. AODV incorporates two key optimizations to localize discovery and repair overhead:

- **Expanding Ring Search (ERS):** Instead of initially broadcasting an RREQ across the entire network diameter, the source sets the IP Time-To-Live (TTL) field to a small initial value (TTL_START, typically 1 or 2 hops). If no RREP arrives within the ring traversal timeout, the source increments TTL by TTL_INCREMENT and rebroadcasts the RREQ. Only if TTL exceeds TTL_THRESHOLD does AODV perform a full network-wide broadcast (TTL = NET_DIAMETER). This rapidly locates nearby destinations with minimal control overhead.
- **Local Route Repair vs. Source-Initiated Rediscovery:** When a link breaks along an active multi-hop route, instead of immediately sending an RERR all the way back to the source, an upstream intermediate node close to the destination can attempt Local Repair. It buffers incoming data packets and broadcasts an RREQ with a localized TTL targeting the destination. If a replacement path is discovered within the discovery period, packet flow resumes without burdening the source node; otherwise, an RERR is dispatched to the source.
- **Link-Layer Feedback vs. Periodic Hello Beacons:** Active nodes monitor neighbor connectivity either via MAC-layer link-failure notifications (such as missing IEEE 802.11 ACKs) or by periodically broadcasting local Hello messages (an RREP with TTL = 1) every HELLO_INTERVAL (typically 1 second). If a node fails to receive any packet or Hello message from an active neighbor for ALLOWED_HELLO_LOSS × HELLO_INTERVAL, it declares the link broken.

8. DSDV versus AODV

Feature	**DSDV (Proactive)**	**AODV (Reactive)**
Route discovery	Routes maintained continuously.	Route found when required.

Routing table	Regularly maintained with destination information.	Maintains routes needed for active communication.
Control overhead	Periodic update messages.	Mainly route-request and route-maintenance messages.
Initial delay	Usually low if route is already available.	May be higher while route discovery occurs.
Mobility	Frequent topology changes trigger table updates.	Broken active routes trigger repair or rediscovery.
Infrequent traffic	May waste bandwidth maintaining unused routes.	Can be efficient because routes are discovered on demand.
Example use	Networks needing ready routes and frequent communication.	Networks with intermittent traffic and changing topology.

9. Reactive Protocol Comparison: AODV versus DSR (Dynamic Source Routing)

Within the class of reactive MANET protocols, AODV is frequently contrasted with Dynamic Source Routing (DSR):

- **Packet Header Overhead:** DSR uses explicit source routing, embedding the complete ordered list of intermediate node IP addresses inside every data packet header. AODV uses hop-by-hop routing tables where data packets carry only the destination IP address, resulting in substantially lower data-packet header overhead over long paths.
- **Route Caching vs. Sequence-Number Freshness:** DSR enables aggressive route caching, allowing nodes to overhear neighboring transmissions (promiscuous mode) and store multiple alternative routes to a destination without timers. However, in high-mobility

scenarios, DSR caches frequently retain stale routes, whereas AODV relies on destination sequence numbers and active route expiration timers to guarantee route freshness.

10. Security Vulnerabilities in Reactive Routing (Blackhole and Rushing Attacks)

Because standard AODV trusts intermediate nodes to faithfully increment hop counts and report genuine sequence numbers, malicious nodes can disrupt reactive route discovery:

- **Blackhole and Grayhole Attacks:** A malicious node replies immediately to every observed RREQ with a forged RREP containing an artificially high Destination Sequence Number and a Hop Count of 1, tricking the source into routing all data traffic through the attacker, which then drops the packets instead of forwarding them.
- **Rushing Attacks:** An attacker forwards RREQ messages immediately without obeying standard MAC-layer jitter delays, ensuring its forged or attacker-controlled path reaches the destination first and suppresses legitimate RREQ broadcasts.

11. Joint Question - Highly Dynamic Network

For a highly dynamic mobile network with frequent topology changes, AODV is generally a suitable choice when traffic is intermittent or bandwidth is limited. It discovers routes only when needed, so it avoids continuously updating every route. When an active link fails, it uses route maintenance and can start a new discovery.

However, AODV has route-discovery delay when a route is not already known. If the application sends frequent traffic and requires very low initial delay, DSDV may be preferred because routes are maintained in advance. The final choice depends on traffic rate, mobility, bandwidth and delay requirements.

Conclusion: AODV is often suitable for dynamic networks with on-demand communication, while DSDV is useful when immediate route availability is more important than periodic control overhead.

Q7 - Design a Mobile Emergency Communication Network

Scenario

During a natural disaster, ambulances, police vehicles, rescue teams and medical staff move continuously. Fixed communication infrastructure may be unavailable. The network must provide reliable communication with minimum delay.

Stage 1 - Independent Design Proposals

Student A - Architecture and Mobility Proposal

- Select an ad hoc wireless network as the primary local communication architecture because fixed infrastructure may be unavailable.
- Treat ambulances, police vehicles and rescue-team devices as mobile nodes.
- Use multi-hop wireless links so nodes can forward messages for other nodes.
- Where a gateway or external network is available, use Mobile IP to maintain reachability as a mobile node changes its network attachment.
- Use fast detection of a new network, care-of address acquisition and registration with the Home Agent.

Student B - Routing and Failure Proposal

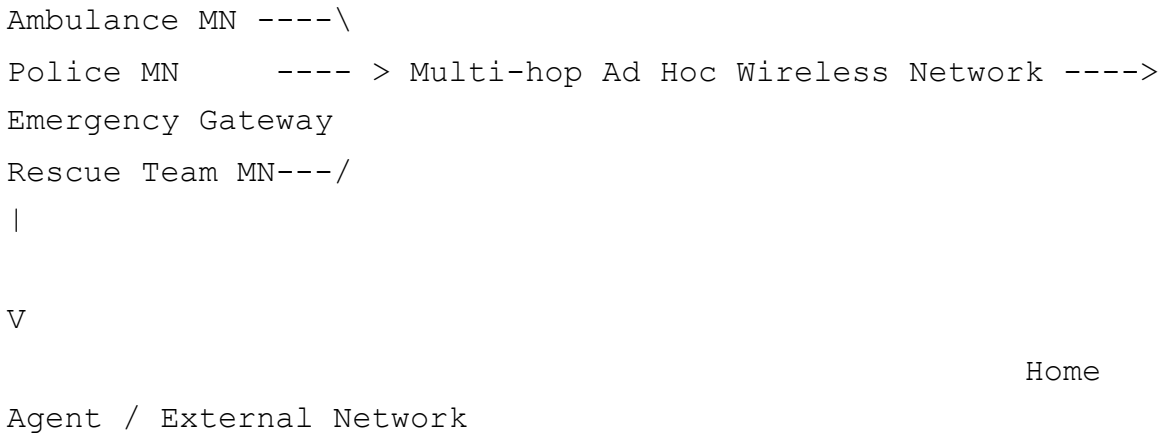
- Use AODV as the main routing protocol for the changing ad hoc network because it discovers routes on demand.
- Use RREQ for route discovery, RREP for route establishment and RERR for link-failure notification.
- Monitor active links and use an alternate route if available.
- If no route is available, start a new route discovery when communication is still required.

- Evaluate packet delivery ratio, end-to-end delay, routing overhead and route recovery time.

Stage 2 - Compare the Two Designs

Area	Student A proposal	Student B proposal	Joint decision
Architecture	Ad hoc local network with gateways where available.	Routing between mobile nodes.	Use a multi-hop ad hoc network.
Mobility	Mobile IP at the boundary with external networks.	Route updates as nodes move.	Use Mobile IP for inter-network mobility and routing for local paths.
Routing	Focus on connectivity architecture.	AODV on demand.	Select AODV for dynamic local routes.
Failure handling	Detect loss of network attachment.	RERR and alternate route discovery.	Use both network reattachment and route repair.
Performance	Reduce handoff interruption.	Measure delay and delivery.	Measure delay, packet delivery and routing overhead.

Stage 3 - Joint Final Network Design



```
    |
    V
    Hospital/Control Centre (CN)
```

Local rescue teams exchange messages through the ad hoc network. When a gateway to an external network is available, the mobile node can use Mobile IP. The Mobile Node obtains a care-of address in the visited network and registers it with the Home Agent. The Home Agent tunnels packets addressed to the home address toward the care-of address.

Mobile IP Registration and Tunnelling

1. The mobile node detects a visited network.

2. It discovers an agent or obtains a care-of address.

3. It sends a registration request to the Home Agent.

4. The Home Agent authenticates the request and stores the home-address/care-of-address binding.

5. Packets from the Correspondent Node reach the Home Agent.

6. The Home Agent encapsulates and tunnels packets to the care-of address.

7. The tunnel endpoint decapsulates and delivers packets to the mobile node.

Selected Routing Protocol: AODV

AODV is selected for local ad hoc communication because the nodes move and the topology changes. AODV discovers routes only when data must be sent. RREQ searches for a path, RREP

establishes the path, and RERR reports a broken route. This reduces continuous routing updates when communication is intermittent.

Failure Handling

Failure	**Action**
Mobile node moves	Update neighbor information and discover or repair a route.
Wireless link fails	Mark the route invalid; use an alternate route or send RERR.
No route exists	Start a new AODV route discovery.
Gateway becomes unavailable	Continue local ad hoc communication; use another available gateway if present.
Mobile IP registration fails	Retry registration and maintain local communication where possible.
Temporary disconnection	Buffer important data where supported and retransmit when connectivity returns.

Gateway Discovery, Buffer Management, and Multi-Channel Emergency Spectrum Allocation

To ensure seamless interoperability between the local multi-hop AODV disaster mesh and the external Mobile IP wide-area hospital network, the joint architecture incorporates three operational mechanisms:

- **Hybrid Emergency Gateway Discovery:** Rescue vehicles and ambulances dynamically locate active Emergency Gateways using hybrid gateway discovery. Gateways periodically broadcast Gateway Advertisement (GWADV) messages with a limited hop radius (TTL = 3),

while outer perimeter nodes requiring external hospital reachability broadcast a Gateway Solicitation (GWSOL) message to discover the nearest functioning gateway.

- **Priority-Aware Handoff Packet Buffering:** During brief intervals when a moving ambulance transitions between two Emergency Gateways or experiences temporary shadow fading, the Mobile IP Foreign Agent and AODV routing layer maintain a prioritized FIFO packet buffer (giving strict priority to vital-sign telemetry and voice dispatch frames over bulk imagery) so packets are retransmitted immediately upon binding renewal rather than dropped.
- **Orthogonal Channel Assignment for Gateway Backhaul:** To prevent intra-flow interference between local multi-hop rescue team communications and high-throughput gateway backhaul links, emergency nodes utilize orthogonal non-overlapping radio channels (e.g., 5 GHz UNII bands or dedicated 4.9 GHz public safety spectrum) for gateway uplinks while reserving 2.4 GHz / sub-6 GHz channels for wider-range peer-to-peer ad hoc links.

Stage 4 - Performance Evaluation

Metric	**Meaning**	**How to evaluate**
End-to-end delay	Time taken for a packet to reach its destination.	Measure send time and receive time; calculate the difference.
Packet delivery ratio	Percentage of sent packets received.	(Packets received / packets sent) × 100.
Routing overhead	Control traffic used for routing.	Count RREQ, RREP, RERR and other routing-control messages.
Route recovery time	Time needed to restore a route after failure.	Measure from link-failure detection until a usable route is available.

Handoff interruption	Time communication is interrupted during network change.	Measure the gap in successful packet delivery during handoff.

Suggested Test Cases

Test	**Condition**	**Expected observation**
1	Small number of mobile nodes; low mobility.	Routes are relatively stable and delay is expected to be lower.
2	Increase the number of mobile nodes.	More route requests and control messages may occur.
3	Increase node speed.	Links change more often; route repairs may increase.
4	Break an active link.	AODV should report the failure and discover another route if possible.
5	Disconnect the gateway.	Local ad hoc communication may continue, but external communication may stop.

No measured simulation values are provided in the assignment question. The table above gives expected observations; actual numerical results should be added only after running a simulation or experiment.

Stage 5 - Cross-Defence

Why is AODV appropriate? It is demand-driven, avoids continuous route updates for unused destinations and provides route maintenance when an active link fails. This is useful when mobile nodes frequently change position.

How does Mobile IP maintain communication? It keeps the mobile node's permanent home address and registers the current care-of address with the Home Agent. The Home Agent tunnels packets to the mobile node's current location.

Stage 6 - Joint Evaluation

- The ad hoc architecture is suitable because it can operate without fixed base stations.
- Mobile IP is useful when a node must remain reachable while moving between different IP networks; it is not a replacement for ad hoc routing inside the local network.
- AODV is selected for dynamic, on-demand local routing.
- The design introduces routing-control messages and Mobile IP registration/tunnelling overhead.
- High mobility can increase delay, packet loss and route-repair frequency.
- When connectivity is lost, local communication may continue if peer links remain; external communication may be unavailable until a gateway or route returns.
- The solution may be unsuitable when node density is too low, wireless links are unreliable, bandwidth is extremely limited or very strict delay guarantees cannot be met.

Final Conclusion

Before selecting Mobile IP and a routing protocol, a network designer should analyse infrastructure availability, node mobility, network size, traffic rate, bandwidth, delay requirement, packet delivery, security, energy limits and failure conditions. Mobile IP manages

movement between IP networks, while routing protocols find paths between nodes. Both should be selected according to the needs of the application.